

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the function $f(x) = \sqrt{1-x}$ on the interval $[-9, -1]$.
 - a) Approximate the area under the curve using a midpoint estimation with 4 subintervals.

 - b) Approximate the area under the curve using a trapezoidal estimation with 4 subintervals.

 - c) Approximate the area under the curve using Simpson's rule with 4 subintervals.

- d) The approximation made in part (a) has some error associated with it which can be estimated by the inequality $|E_M| \leq \frac{K(b-a)^3}{24n^2}$ where $|f''(x)| \leq K$ for all x in $[a, b]$.
- i) Determine a value of K which can be used to help estimate the upper bound on the error.

- ii) Determine the upper bound on the error of the approximation from part (a).

- e) Determine the values of n needed for a midpoint approximation and trapezoidal approximation to estimate the total area under $f(x)$ on $[-9, -1]$ to within $\frac{1}{100}$.

2. Determine the value of n needed for Simpson's rule to estimate the area under a curve $f(x)$ on $[-1, 1]$ to within $\frac{1}{100}$ if we know that $f^{(4)}(x) = 3x^{1/3} - 6x^{4/3}$.

3. Does the improper integral $\int_1^{\infty} \sin(x) dx$ converge or diverge? Why?

4. Determine whether the integral is convergent or divergent. If convergent, evaluate the integral.

a) $\int_0^1 \frac{1}{\sqrt{x}} dx$

b) $\int_0^{\infty} \frac{x^2}{27+x^3} dx$